

TATEYAMA AB, Japan

WMO: 476880

Lat: 34.987N

Lon: 139.832E

Elev: 6

StdP: 101.25

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	0.1	1.2	-10.1	1.6	4.9	-8.8	1.8	5.2	16.7	9.0	14.9	9.1	2.3	130	0.446

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	5.6	31.2	26.3	30.3	26.0	29.8	25.8	27.0	29.8	26.6	29.2	26.2	28.6	4.6	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.2	21.6	28.8	26.0	21.3	28.7	25.2	20.3	27.9	85.5	30.2	83.5	29.5	81.8	28.7	28.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.9	10.9	9.2	-2.1	33.2	0.8	1.1	-2.7	34.0	-3.1	34.6	-3.6	35.2	-4.2	36.0		
(5)			-4.0	27.8	0.8	0.6	-4.5	28.2	-5.0	28.6	-5.4	28.9	-6.0	29.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	16.6	6.9	7.4	10.1	14.5	18.6	21.6	25.5	27.1	24.0	19.2	14.1	9.7	(6)
(7)		DBStd	7.28	2.38	2.70	3.00	3.04	2.23	2.03	2.22	1.83	2.69	2.62	2.80	2.65	(7)
(8)		HDD10.0	262	101	83	36	3	0	0	0	0	0	0	2	37	(8)
(9)		HDD18.3	1474	353	307	254	118	24	1	0	0	1	21	128	267	(9)
(10)		CDD10.0	2679	6	9	40	138	267	349	481	530	421	284	127	28	(10)
(11)		CDD18.3	850	0	0	0	3	32	101	223	272	171	46	3	0	(11)
(12)		CDH23.3	6033	0	0	0	1	40	281	1711	2675	1190	134	1	0	(12)
(13)		CDH26.7	1463	0	0	0	0	1	17	397	798	244	7	0	0	(13)
(14)	Wind	WSAvg	4.0	4.7	4.5	4.7	4.3	3.9	3.6	3.8	3.7	3.5	3.6	3.8	4.5	(14)
(15)	Precipitation	PrecAvg	1917	90	99	163	166	171	221	176	122	229	236	154	90	(15)
(16)		PrecMax	2382	228	189	293	320	304	424	456	346	463	646	472	258	(16)
(17)		PrecMin	1268	1	19	57	83	85	88	4	2	68	51	44	5	(17)
(18)		PrecStd	303	55	47	59	61	56	88	122	91	101	142	94	57	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.0	18.1	19.9	22.9	25.9	28.7	32.1	32.8	31.2	27.8	23.0	20.2	(19)
(20)			MCWB	14.2	14.9	15.2	18.1	20.0	24.0	26.1	26.8	25.8	22.9	18.7	17.4	(20)
(21)		2%	DB	15.0	16.2	18.2	21.2	24.2	27.0	31.0	31.8	30.1	26.0	21.8	18.1	(21)
(22)			MCWB	11.4	13.1	14.6	17.3	19.7	22.9	26.2	26.4	25.7	22.4	18.2	14.5	(22)
(23)		5%	DB	13.2	14.4	17.1	20.2	23.2	25.9	30.0	31.0	29.1	24.9	20.8	16.2	(23)
(24)			MCWB	8.9	10.9	13.9	16.4	19.0	22.4	25.8	26.4	25.1	21.3	17.1	12.0	(24)
(25)		10%	DB	12.0	12.9	16.0	19.2	22.8	25.0	29.0	30.1	28.1	23.8	19.2	14.9	(25)
(26)			MCWB	7.6	8.9	12.4	15.6	18.8	22.0	25.3	26.1	24.6	20.2	15.5	10.6	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.0	15.7	17.3	20.4	22.6	25.2	27.2	27.8	27.1	24.7	20.6	18.3	(27)
(28)			MCD B	16.6	17.4	18.5	21.4	23.9	27.4	30.0	30.4	29.3	26.1	22.0	19.7	(28)
(29)		2%	WB	12.1	13.8	15.6	18.6	21.3	24.3	26.7	27.2	26.4	23.3	18.8	15.1	(29)
(30)			MCD B	14.5	15.9	17.6	20.6	23.1	26.0	29.5	30.0	28.6	25.2	20.8	17.3	(30)
(31)		5%	WB	9.8	11.4	14.5	17.5	20.4	23.5	26.2	26.9	25.9	21.8	17.5	12.8	(31)
(32)			MCD B	12.6	14.0	16.8	19.8	22.5	25.1	28.8	29.6	28.1	24.2	19.8	15.7	(32)
(33)		10%	WB	7.9	9.4	13.0	16.5	19.6	22.6	25.8	26.5	25.4	20.6	16.3	11.1	(33)
(34)			MCD B	10.9	12.1	15.5	18.8	21.7	24.4	28.1	29.1	27.6	23.0	18.8	14.0	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	7.7	7.6	7.5	7.4	6.7	5.5	5.3	5.6	6.0	6.4	7.2	7.5	(35)
(36)			MCDBR	9.2	9.2	8.4	8.0	7.8	6.4	6.5	6.3	6.3	7.2	8.3	8.9	(36)
(37)			MCWBR	7.5	7.9	6.8	5.4	4.5	3.6	3.1	2.8	3.4	4.9	6.1	7.1	(37)
(38)		5% WB	MCD BR	8.6	8.6	7.6	6.5	6.2	5.0	5.7	5.6	5.6	6.2	6.9	8.4	(38)
(39)			MCWBR	7.7	8.1	7.3	5.4	4.6	3.4	3.0	2.8	3.4	5.2	6.0	7.5	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.388	0.438	0.500	0.543	0.557	0.573	0.568	0.551	0.500	0.483	0.432	0.389	(40)	
(41)		taud	2.234	2.091	1.942	1.882	1.910	1.947	2.003	2.044	2.158	2.120	2.179	2.238	(41)	
(42)		Ebn at Noon	791	788	770	759	754	740	740	744	764	734	734	762	(42)	
(43)		Edh at Noon	109	140	177	197	194	187	176	166	141	135	115	103	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.63	3.12	3.98	4.70	5.12	4.50	5.17	5.24	4.03	3.02	2.45	2.33	(44)	
(45)		RadStd	0.21	0.35	0.28	0.49	0.54	0.45	0.84	0.44	0.35	0.32	0.22	0.20	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (36 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page